



Lab 3: Paradigms blender

Overview

In this assignment, you will design and implement a **Smart Study Advisor** system using multiple programming paradigms. The goal is to understand how different paradigms contribute to building intelligent systems.

You will build **two versions** of the system:

- An **AI-based system** using external APIs
- A **non-AI system** using programming paradigms

Team Structure

Each team consists of **4 students**. Each member must be able to demonstrate all the work.

1. Logic Programmer (Prolog)

- Implement the inference engine using Prolog
- Define rules such as:
 - Course difficulty
 - Prerequisites
 - Student preferences
- Provide queries such as:

```
recommend(Student, Course) .
```

2. Backend Engineer

- Implement a Django server
- You should demonstrate where each paradigm shines (OOP, Imperative, Functional)
- Integrate with Prolog engine

System Architecture

The non-AI system must follow this pipeline:

Mobile App → Django Backend → Prolog Inference Engine → Django → Mobile App

Mobile Application

- Simple interface:
 - Input preferences
 - Display recommendations
- Any framework is allowed (Flutter, React Native, etc.)
- **Note:** Mobile code is allowed to be vibecoded

Part 1: AI-Based Advisor

- Integrate an external AI API (e.g., Gemini)
- Send structured prompts
- Display recommendations from the model

Part 2: Non-AI Advisor

- Must use all paradigms:
 - Logic (Prolog) → reasoning
 - Functional → data transformation
 - OOP → system design
 - Imperative → execution flow

Deliverables

1. Code

- Github repo link with no commits after deadline
- Repo should contain a README file describing your work

2. Report

Include the following sections:

- **Architecture:**
 - Where each paradigm is used
 - Why it fits there ?

- **Reflection:**

If you were building a real AI system, what approach would you choose and why?

3. Demo

- Show both versions working (Each video 2 – 3 minutes)

Guidelines and Submission Details

- No Late Policy, so try not to miss the discussion
- Lab deadline is on Wednesday 13th. Use the following [form](#) to submit your solution.